things. First, it would provide direct evidence of an ancient era when various aspects of today's universe that appear distinct were part of a symmetric whole. Second, it would establish that our intuitive notion of empty space—the end result of removing everything we can from a region so that its energy and temperature drop as low as possible—has, for a long time, been naïve. The emptiest empty space need not involve a state of absolute nothingness. Without invoking the spiritual, therefore, we may well closely brush up against the thinking of Henry More (Chapter 2) in our scientific quest to understand space and time. To More, the usual concept of empty space was meaningless because space is always filled with divine spirit. To us, the usual concept of empty space may be similarly elusive, since the empty space we're privy to may always be filled with an ocean of Higgs field.

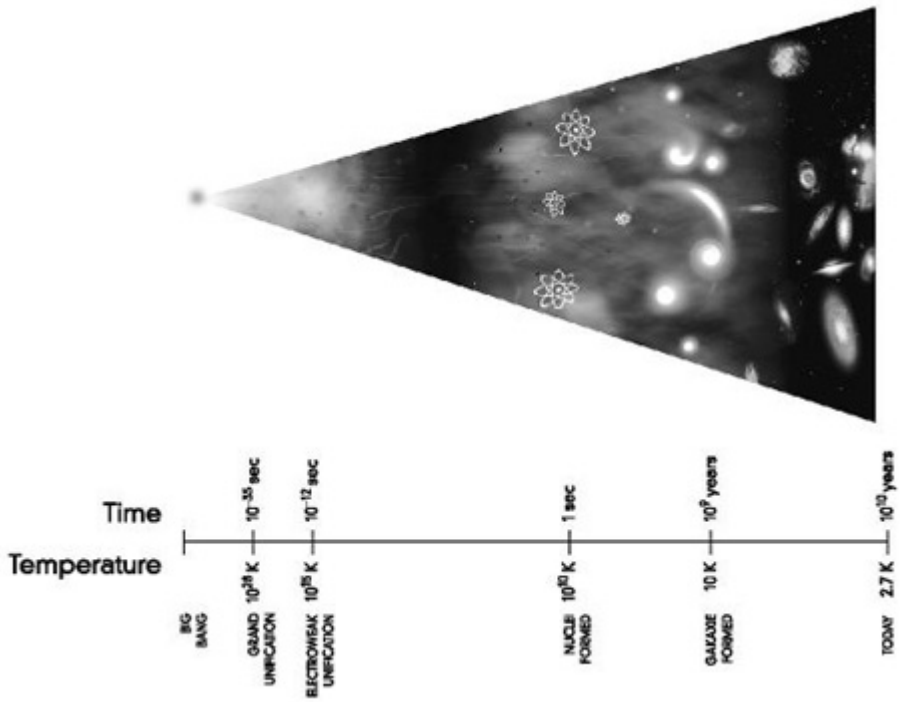


Figure 9.2 A time line schematically illustrating the standard big bang model of cosmology.

Entropy and Time

The time line in Figure 9.2 places the phase transitions we've discussed in historical context and hence gives us a firmer grasp of the sequence of events the universe has gone through from the big bang to the egg on